



Fabric that can regulate heat

The new fabric engineered using synthetic yarn with a carbon nanotube coating can automatically cool or insulate <https://digit.in/mar19-15>



CRISPR babies are enhanced

The genetically edited twin girls in China may have enhanced memory and cognition. <https://digit.in/mar19-16>

2D Materials

Sheets of materials with the thickness of a single atom can revolutionise electronics, but commercial application remains a challenge

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In 2004, two researchers at Manchester University developed a habit of meeting on Friday nights, and conducting experiments that had nothing to do with their actual work. During one of these sessions, Professor Andre Geim and Professor Kostya Novoselov stuck pieces of tape to an ordinary block of graphite, the kind used in pencils, and ripped it off. This would turn out to be a groundbreaking exercise. After repeatedly doing this, they managed to isolate a tiny slice of carbon. Although scientists knew that a form of carbon existed, which had a thickness of a single atom, no one had managed to isolate it. This was a single molecule made up of a hexagonal honeycomb lattice of carbon atoms, known as graphene.

Graphene turned out to have some pretty interesting properties. It was stronger than steel, conducted electricity as well as copper and conducted heat better than any other known material. It was also almost entirely transparent but at the same time so dense that even the smallest atom, helium, cannot

pass through it. The two scientists were rewarded for goofing around in their spare time, with the Nobel Prize for Physics in 2010. "Playfulness is one of their hallmarks, one always learns something in the process and, who knows, you may even hit the jackpot. Like now when they, with graphene, write themselves into the annals of science," the prize citation notes.

Most of the 2D materials are crystalline allotropes (different physical forms of the same element) that are either one atom thick, or a few atoms thick. For example, silicene is made up of silicon, and has a hexagonal honeycomb lattice structure like graphene. While 2D materials are well understood, there are few practical applications at scale. The properties of graphene are ideally suited for various applications in electronics, including touchscreens, displays and solar panels. However, integration with electronic systems has remained a challenge, which is expected to be overcome in the near future. Of all the 2D materials known, graphene is the closest to being used in widespread commercial applications.

RESEARCH

Since the isolation of graphene, over 700 new 2D materials have been

discovered, out of thousands that are predicted to exist. However, only about 50 materials have been physically made in laboratories. Finding new 2D materials is a field of active research, and there are new materials being discovered all the time. This kind of research takes place in India as well. In December 2018, researchers from the Cochin University of Science and Technology, discovered three new 2D materials after collaborating with international universities. These are hematene, ilmenite and manganese telluride, derived from minerals commonly available in Kerala. The 2D materials can split water into hydrogen and oxygen, and have potential applications in electronics.

Some promising materials are known theoretically, but have not yet been synthesized. An example of such a material is graphyne, which is similar to graphene, and expected to be a replacement in the future. Scientists can work out how elements can form molecules that are 2D sheets, and predict the existence of 2D materials before they are found or synthesized in laboratories. Supercomputers are employed to simulate various extreme conditions, to investigate the properties of potential 2D materials.